Pakistan Air Intelligence

Multi-City Air Quality Forecasting Using Machine Learning & Deep Learning

Software Requirements Specification (SRS)

Field	Detail
Document Version	1.0
Project Type	Machine Learning + Deep Learning + Time-Series Forecasting
Dataset	Pakistan Air Quality and Weather — 10 Cities (Kaggle)
Primary Goal	Predict next-hour PM2.5 and classify AQI category across 10 Pakistani cities using historical air-quality, weather and time-based features, while comparing ML and DL model performance.

1. Introduction

1.1 Purpose

The purpose of this project is to develop an intelligent air-quality prediction system that analyzes air pollution patterns across multiple Pakistani cities and forecasts future air quality using historical pollution, weather, and time-based features.

The system will perform two primary predictive tasks:

1. PM2.5 Regression — predict the next-hour PM2.5 concentration.
2. AQI Classification — classify air quality into predefined AQI categories.

Both traditional Machine Learning and Deep Learning models will be implemented, and their performance will be compared.

2. Problem Statement

Air pollution is a major environmental problem in Pakistan's urban areas. Pollution levels change with weather conditions, traffic, industrial activity, season, and time of day.

Traditional monitoring only provides current air quality. The objective of this project is to learn from historical data patterns and provide future air-quality forecasting.

Central Question

> *“Given the current and historical air-quality, weather, time and city information, what will the PM2.5 concentration be during the next hour?”*

3. Project Objectives

Primary Objectives

3. Clean and preprocess the air-quality dataset.
4. Handle missing values and inconsistent records.
5. Perform Exploratory Data Analysis (EDA).
6. Identify relationships between pollution and weather.
7. Generate time-series features.
8. Train ML models for PM2.5 prediction.
9. Develop AQI category classification models.
10. Implement ANN, LSTM and GRU models.
11. Compare ML vs DL performance.
12. Identify the best-performing model.
13. Provide model explainability.
14. Save the final model in a deployable format.

4. Scope

4.1 Included

Data Processing

- Data loading
- Data validation
- Missing-value handling
- Duplicate removal
- Outlier analysis
- Data type conversion
- Date/time processing

Exploratory Data Analysis

- City-wise pollution analysis
- PM2.5 distribution
- PM10 distribution
- AQI distribution
- Weather analysis
- Correlation analysis
- Hourly trends
- Daily trends
- Monthly trends
- City comparison

Feature Engineering

- Hour
- Day
- Day of week
- Month
- Weekend indicator
- Season
- Lag features
- Rolling averages

Machine Learning — Regression

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting
- XGBoost

Machine Learning — Classification

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

- XGBoost

Deep Learning

- Artificial Neural Network (ANN)
- LSTM
- GRU

Evaluation — Regression

- MAE
- MSE
- RMSE
- R^2

Evaluation — Classification

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC where applicable

Explainability

- Feature importance
- SHAP analysis
- Prediction error analysis

5. Out of Scope

The following features will not be included in the initial version:

- Real-time IoT sensor integration
- Satellite imagery
- Weather API integration
- LLM / chatbot
- RAG
- AI agents
- Mobile application
- Automatic government alert system
- Medical diagnosis
- Industrial sensor integration

These features may be added in future versions.

6. Dataset Requirements

Selected dataset: Pakistan Air Quality and Weather — 10 Cities.

The dataset contains approximately:

- 21,840 hourly records
- 10 Pakistani cities
- Air-quality variables
- Weather variables
- Time-related information

Major Data Categories

Category	Variables
Air Quality	PM2.5, PM10, CO, NO_2, SO_2, O_3, Dust
Weather	Temperature, Humidity, Pressure, Wind Speed, Wind Direction, Precipitation, Irradiance/weather-related variables where available
Geographic	City, Latitude, Longitude
Temporal	Date, Time, Hour, Day, Month

7. Functional Requirements

FR-01 — Dataset Loading

The system shall successfully load the CSV dataset.

Input	Output
Dataset CSV	Pandas DataFrame

FR-02 — Data Validation

The system shall automatically check for:

- Missing values
- Duplicate records
- Invalid data types
- Negative pollution values
- Invalid timestamps
- Extreme outliers

FR-03 — Data Preprocessing

The system shall:

- Handle missing values
- Remove duplicate rows
- Convert date/time fields
- Encode categorical features
- Scale numerical features where required

8. Exploratory Data Analysis Requirements

FR-04 — City Analysis

The system shall compare pollution levels for each city, e.g.:

City	Average PM2.5
Lahore	X
Karachi	X
Islamabad	X
Rawalpindi	X
…	…

FR-05 — Temporal Analysis

The system shall analyze pollution levels on an hourly, daily, weekly, and monthly basis.

FR-06 — Correlation Analysis

The system shall identify relationships between air-quality and weather variables, e.g.:

- Temperature ↔ PM2.5
- Humidity ↔ PM2.5
- Wind Speed ↔ PM2.5
- PM10 ↔ PM2.5
- NO_2 ↔ PM2.5

9. Feature Engineering Requirements

FR-07 — Time Features

- Hour
- Day
- Day of Week
- Month
- Weekend
- Season

FR-08 — Lag Features

For time-series forecasting, the system shall generate:

- PM2.5(t-1)
- PM2.5(t-2)
- PM2.5(t-3)
- PM2.5(t-6)
- PM2.5(t-12)
- PM2.5(t-24)

FR-09 — Rolling Features

- 3-hour rolling mean
- 6-hour rolling mean
- 12-hour rolling mean
- 24-hour rolling mean

10. Prediction Tasks

FR-10 — PM2.5 Regression

Input

Current and historical: PM2.5, PM10, CO, NO_2, SO_2, O_3, Temperature, Humidity, Pressure, Wind speed, Time features, City.

Output

Predicted PM2.5 for the next hour.

```
Current PM2.5:    84.2 µg/m³

Predicted next hour:
   91.7 µg/m³
```

11. AQI Classification

FR-11 — AQI Category Prediction

The system shall predict the air-quality category based on pollution information. Possible outputs:

- Good
- Moderate
- Unhealthy for Sensitive Groups
- Unhealthy
- Very Unhealthy
- Hazardous

AQI category thresholds will be explicitly documented according to the standard chosen at implementation time.

12. Machine Learning Requirements

FR-12 — Regression Models

Type	Models
Baseline	Linear Regression
Tree Models	Decision Tree, Random Forest
Ensemble	Gradient Boosting, XGBoost

Each model's performance shall be evaluated and compared.

FR-13 — Classification Models

- Logistic Regression
- Decision Tree
- Random Forest

- Gradient Boosting
- XGBoost

13. Deep Learning Requirements

FR-14 — ANN

```
Input Layer
     ↓
   Dense
     ↓
   ReLU
     ↓
  Dropout
     ↓
   Dense
     ↓
   ReLU
     ↓
  Output
```

FR-15 — LSTM

Using historical time windows, the LSTM shall predict the next-hour PM2.5.

```
Past 24 Hours
      ↓
    LSTM
      ↓
Next Hour PM2.5
```

FR-16 — GRU

The GRU model shall perform the same forecasting task and be compared against the LSTM.

14. Model Training Requirements

The system shall:

- Create train / validation / test datasets
- Use chronological splitting for time-series data
- Prevent data leakage
- Perform feature scaling where required
- Support hyperparameter tuning
- Save the best model

Recommended Split

Set	Percentage
Training	70%
Validation	15%
Testing	15%

Chronological order shall be maintained across all splits.

15. Model Evaluation

15.1 Regression

Metric	Description
MAE	Average prediction error
MSE	Large errors are given a higher penalty
RMSE	Prediction error represented in original target units
R^2	Measures the model's explained variance

15.2 Classification

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

16. Model Comparison

The final system shall automatically generate a comparison table, e.g.:

Model	MAE	RMSE	R^2
Linear Regression	—	—	—
Decision Tree	—	—	—
Random Forest	—	—	—
Gradient Boosting	—	—	—
XGBoost	—	—	—
ANN	—	—	—
LSTM	—	—	—
GRU	—	—	—

The system shall identify the best regression model based on the lowest RMSE/MAE and the highest R^2.

17. Visualization Requirements

EDA

- Histogram
- Box plot
- Correlation heatmap
- Line plots
- City comparison
- Distribution plots

Forecasting

- Actual vs Predicted
- Prediction error
- Residual plot
- Future PM2.5 trend

Model Comparison

- RMSE comparison
- MAE comparison
- R^2 comparison

Deep Learning

- Training loss
- Validation loss
- Training metrics
- Validation metrics

18. Explainable AI

Feature importance shall be generated for the best-performing ML model, e.g.:

Feature	Relative Importance
PM2.5 Lag 1	██████████
PM10	████████
NO2	██████
Humidity	█████
Temperature	████
Wind Speed	███

Optional advanced implementation: SHAP, to explain individual predictions.

Example: a high PM2.5 prediction is primarily driven by the previous-hour PM2.5, PM10 and NO_2.

19. Streamlit Application

The final system may be deployed as a Streamlit application.

Dashboard — Home

```
   Pakistan Air Intelligence

Multi-City Air Quality
ML + Deep Learning Forecasting
```

Prediction Workflow

The user selects: City, Date/Time, Weather Parameters, Pollution Parameters.

```
Predicted PM2.5
       ↓
  AQI Category
       ↓
  Risk Level
```

20. Dashboard Pages

Page	Description
Page 1 — Overview	Dataset statistics, cities, average PM2.5, average PM10, AQI overview
Page 2 — EDA	Pollution analysis, weather analysis, correlation, city comparison
Page 3 — PM2.5 Prediction	Input values → predicted PM2.5
Page 4 — AQI Classification	Input values → AQI category
Page 5 — Model Comparison	ML vs DL performance
Page 6 — Time-Series Forecast	Historical → future PM2.5 trend
Page 7 — Explainability	Feature importance + SHAP

21. Non-Functional Requirements

ID	Requirement	Description
NFR-01	Performance	Predictions should ideally return within a few seconds after model loading
NFR-02	Reliability	Invalid user inputs shall be properly validated
NFR-03	Usability	The UI shall be simple and understandable
NFR-04	Reproducibility	Random seeds and the preprocessing pipeline shall be documented
NFR-05	Maintainability	The project shall be developed in a modular structure

22. Technology Stack

Category	Tools / Technologies
Programming	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost
Deep Learning	TensorFlow / Keras
Explainability	SHAP
Deployment	Streamlit
Development	Jupyter Notebook, VS Code / PyCharm, Git & GitHub

23. Hardware Requirements

Recommended

- Intel Core i5 / Ryzen 5 or better
- 8 GB RAM minimum
- 16 GB RAM recommended
- SSD recommended

GPU

Optional. ANN/LSTM/GRU training is feasible on a normal CPU given the dataset size, although GPU training will be faster.

24. Expected Outputs

#	Output
1	Next-hour PM2.5 prediction
2	AQI category
3	Best ML model
4	Best DL model
5	ML vs DL comparison
6	City-wise air-quality analysis
7	Time-series forecasting
8	Feature importance / explainability

25. Success Criteria

The project shall be considered successful if:

- The dataset is successfully processed
- No data leakage occurs
- Multiple ML models are compared
- ANN, LSTM and GRU are successfully trained
- Regression metrics are calculated
- AQI classification metrics are calculated
- The best model is objectively selected
- Actual vs predicted visualization is available
- City-wise analysis is available
- The final model is usable in Streamlit

26. Future Enhancements

```
Real-Time Sensors
       ↓
   Weather API
       ↓
  Real-Time AQI
       ↓
 Live Forecasting
       ↓
Automated Alerts
```

Further:

- Satellite data
- More Pakistani cities
- Longer historical dataset

- Transformer-based time-series model
- Real-time monitoring
- Mobile application

However, Version 1 will deliberately exclude these features so the project's core focus remains on Machine Learning + Deep Learning + Time-Series Forecasting.

27. Final Project Definition

> **Pakistan Air Intelligence**
> A multi-city Machine Learning and Deep Learning based air-quality forecasting system that analyzes historical pollution, weather and temporal data from 10 Pakistani cities to predict next-hour PM2.5 concentration and classify air-quality categories, while systematically comparing traditional ML models with ANN, LSTM and GRU deep-learning models.

Core Concepts Covered

EDA → Data Cleaning → Feature Engineering → Regression → Classification → Ensemble ML → Time Series → ANN → LSTM → GRU → Hyperparameter Tuning → Model Evaluation → Explainable AI → Deployment